

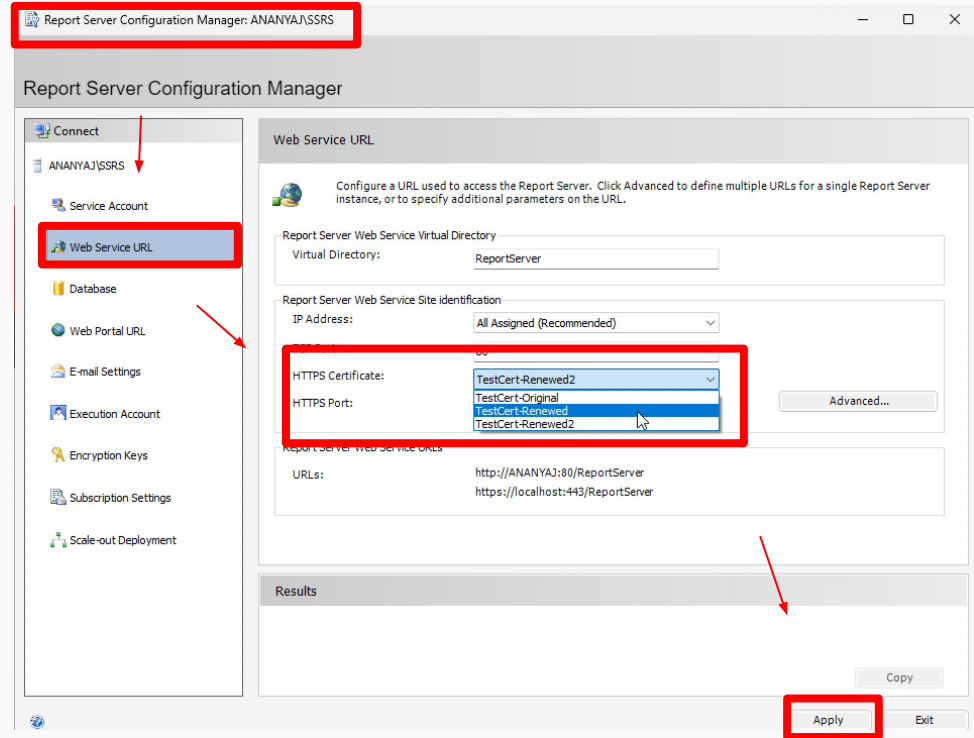
Automating SSL Certificate Binding for SSRS



Current Procedure

After certificate renewal (either manual or automated via win-acme):

1. Open 'Report Server Configuration Manager'
2. Go to 'Web Service URL'
3. Select renewed certificate from 'HTTPS Certificate Dropdown'
4. Click 'Apply' to create new certificate binding



Issues with Current Procedure

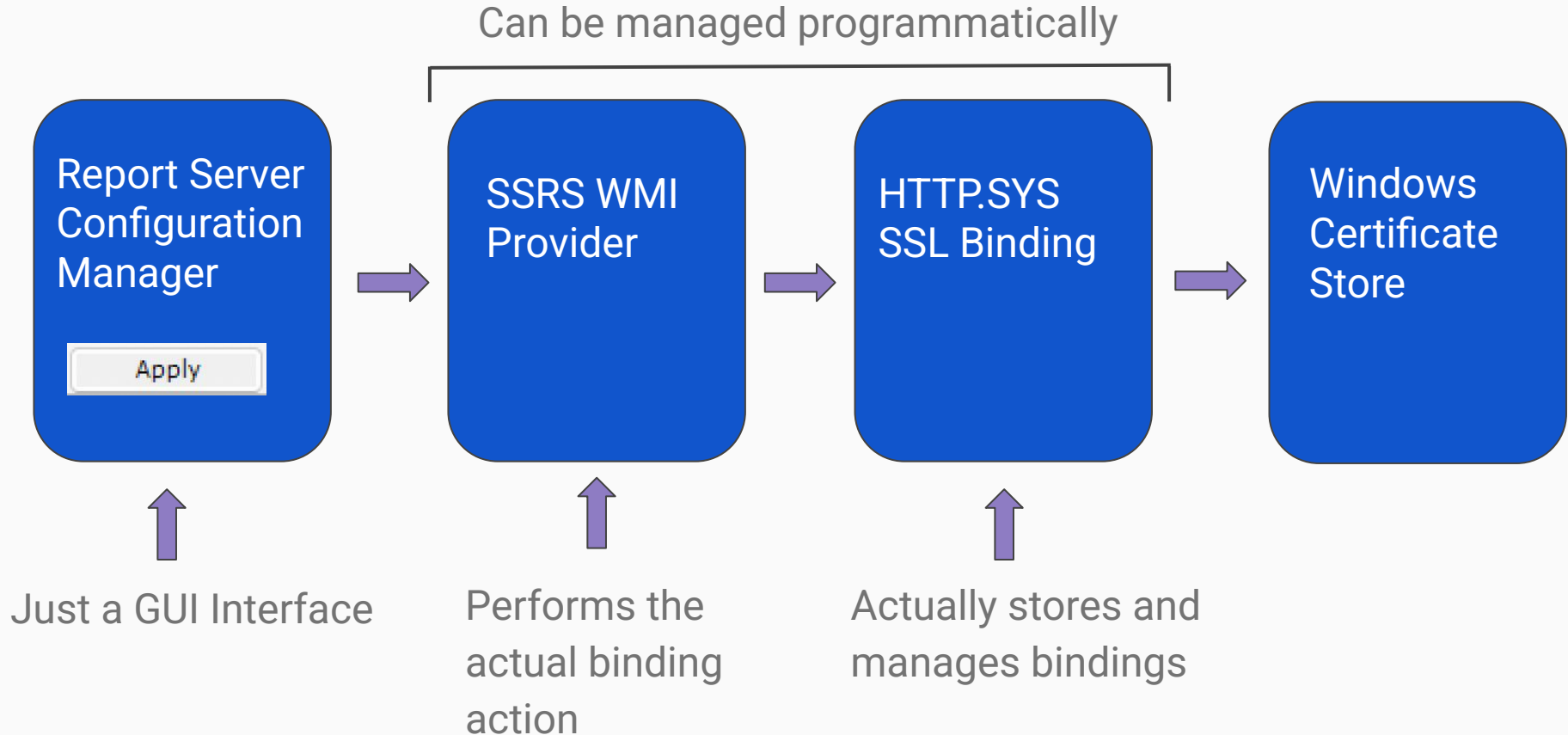
Operational Risk:

1. **Time consuming:** requires manual navigation through Report Server Configuration Manager
2. **Doesn't scale for many servers:** must be done per renewal, per server
3. **Possibility of Human Error:** in case incorrect certificate is selected

Reliability Concerns:

1. **Single Point of Failure:** relies on availability of administrator
2. **Downtime risk:** delay between expiry and re-binding makes data inaccessible for some time period

Why SRSS Certificate Bindings Can Be Automated



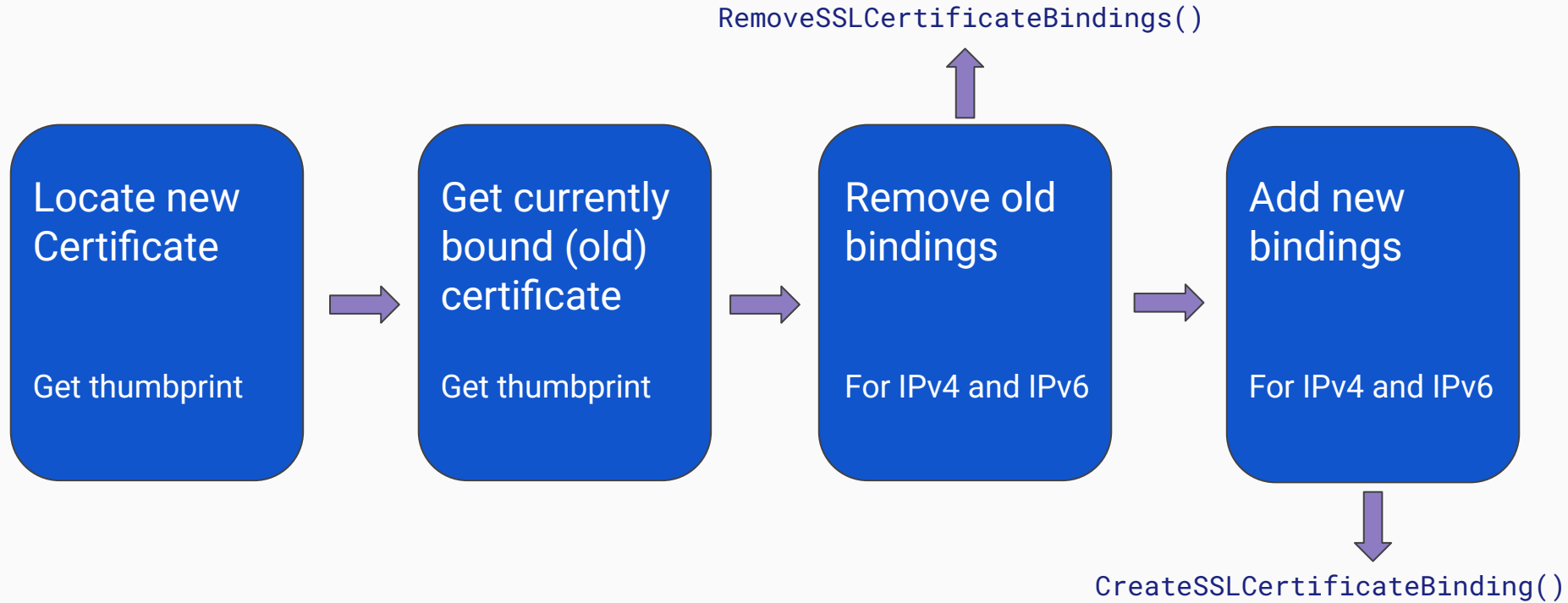
Relevant WMI Operations for SSRS

<u>Method</u>	<u>Description</u>
CreateSSLCertificateBinding Method (WMI MSReportServer_ConfigurationSetting)	Creates a TLS/SSL Certificate binding
ListSSLCertificateBindings Method (WMI MSReportServer_ConfigurationSetting)	Lists TLS/SSL certificate bindings that exist in HTTP.SYS and those bindings expected from rsreportserver.config
ListSSLCertificates Method (WMI MSReportServer_ConfigurationSetting)	Lists installed TLS/SSL certificates on the computer
RemoveSSLCertificateBindings Method (WMI MSReportServer_ConfigurationSetting)	Remove a TLS/SSL certificate binding

Docs Reference:

<https://learn.microsoft.com/en-us/sql/reporting-services/wmi-provider-library-reference/msreportserver-configurationsetting-members?view=sql-server->

Powershell Script Structure



Script I - Testing Initial Feasibility

Lookup Parameter: FriendlyName

```
# Get new certificate using FriendlyName
$newCert = Get-ChildItem Cert:\LocalMachine\My | Where-Object { $_.FriendlyName -eq "TestCert-Renewed" }
$testCertRenewedThumbprint = $newCert.Thumbprint.ToLower() # WMI expects certificate thumbprint in lowercase
Write-Host "TestCert-Renewed thumbprint (lowercase): " $testCertRenewedThumbprint

# Get old certificate using FriendlyName
$oldCert = Get-ChildItem Cert:\LocalMachine\My | Where-Object { $_.FriendlyName -eq "TestCert-Original" }
$testCertOriginalThumbprint = $oldCert.Thumbprint.ToLower()
Write-Host "TestCert-Original thumbprint (lowercase): " $testCertOriginalThumbprint

# Connect to SSRS via WMI
$ssrs = Get-WmiObject -Namespace "root\Microsoft\SqlServer\ReportServer\RS_SSRS\v16\Admin" -Class MSReportServer_ConfigurationSetting

# Remove old certificate bindings
$ssrs.RemoveSSLCertificateBindings("ReportServerWebService", $testCertOriginalThumbprint, "0.0.0.0", 443, 1033) #IPv4
$ssrs.RemoveSSLCertificateBindings("ReportServerWebService", $testCertOriginalThumbprint, ":::", 443, 1033) #IPv6

# Create new certificate bindings
$ssrs.CreateSSLCertificateBinding("ReportServerWebService", $testCertRenewedThumbprint, "0.0.0.0", 443, 1033) #IPv4
$ssrs.CreateSSLCertificateBinding("ReportServerWebService", $testCertRenewedThumbprint, ":::", 443, 1033) #IPv6
```

Script II - More generalizable

Lookup Parameter: domain and **NotBefore** (day certificate becomes valid)

```
# Get new certificate using domain and issue date
$dnsName = "localhost"
$newCert = Get-ChildItem Cert:\LocalMachine\My |
    Where-Object { $_.Subject -like "*CN=$dnsName*" } |
    Sort-Object NotBefore -Descending |
    Select-Object -First 1
$newCertThumbprint = $newCert.Thumbprint.ToLower()
Write-Host "Thumbprint (lowercase): " $newCertThumbprint

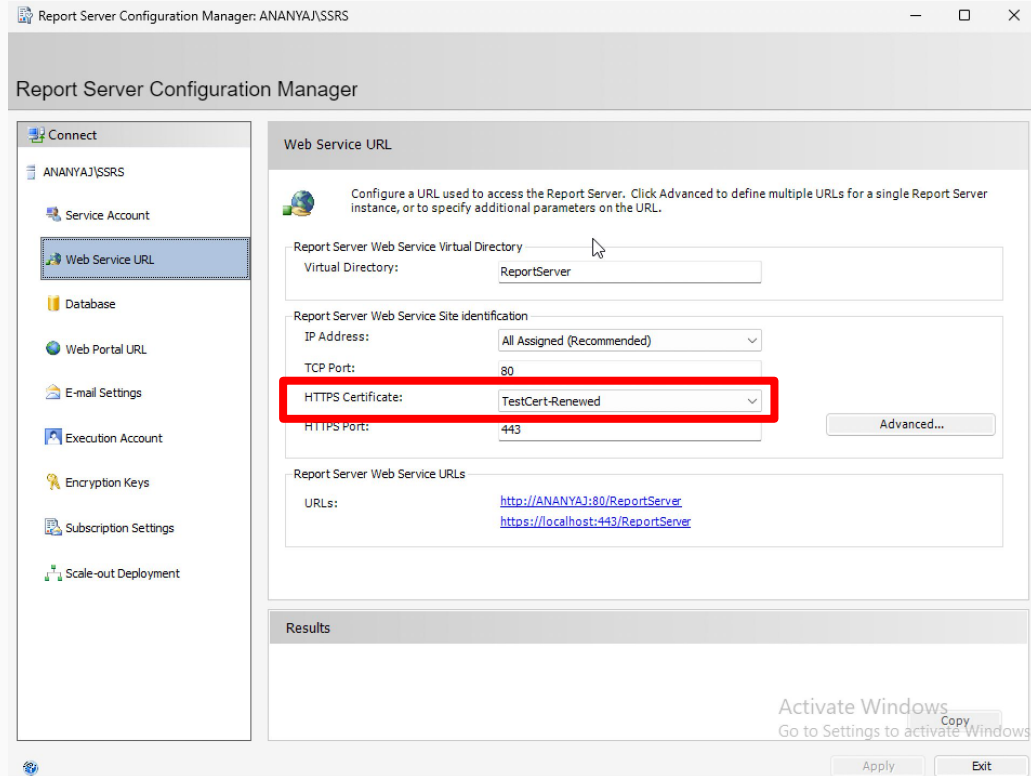
# Get old certificate using name
$soldCert = Get-ChildItem Cert:\LocalMachine\My | Where-Object { $_.FriendlyName -eq "TestCert-Original" }
$soldCertThumbprint = $soldCert.Thumbprint.ToLower()
Write-Host "Currently bound certificate thumbprint: " $soldCertThumbprint

# Connect to SSRS via WMI
$ssrs = Get-WmiObject -Namespace "root\Microsoft\SqlServer\ReportServer\RS_SSRS\v16\Admin" -Class MSReportServer_ConfigurationSetting

# Remove old binding
$ssrs.RemoveSSLCertificateBindings("ReportServerWebService", $soldCertThumbprint, "0.0.0.0", 443, 1033)
$ssrs.RemoveSSLCertificateBindings("ReportServerWebService", $soldCertThumbprint, ":", 443, 1033)

# Add new binding
$ssrs.CreateSSLCertificateBinding("ReportServerWebService", $newCertThumbprint, "0.0.0.0", 443, 1033)
$ssrs.CreateSSLCertificateBinding("ReportServerWebService", $newCertThumbprint, ":", 443, 1033)
```

Auto-Execution of Powershell Script



The binding can be reliably updated using Powershell and SSRS's WMI Provider

Script still requires manual execution



Automating Script Execution

WIN-ACME

WIN-
ACME

A simple ACMEv2 client for Windows (for use with Let's Encrypt et al.)

DOWNLOAD
2.2.9.1

Features:

- Interface to create and install certificates
- Create scheduled tasks to renew certificates
- Store your certificates where and how you want them
- Completely unattended operation from the command line
- **Write your own Powershell .ps1 scripts to handle installation and validation**

Docs Reference: <https://www.win-acme.com>

Docs Reference: <https://www.win-acme.com/reference/plugins/installation/script>

Script

Runs an external script or executable after a succesful renewal. This may be a `.bat`, `.ps1` or even `.exe`. You provide the program with the path to the script and it will run automatically.

Parameters

The following variables can be provided from the program to the script as command line arguments.

Value	Replaced with
<code>{0}</code> or <code>{CertCommonName}</code>	Common name (primary domain name) of the newly issued certificate
<code>{1}</code> or <code>{CachePassword}</code>	The <code>.pfx</code> password (generated randomly for each renewal)
<code>{2}</code> or <code>{CacheFile}</code>	Full path of the cached <code>.pfx</code> file (*)
<code>{6}</code> or <code>{CacheFolder}</code>	Directory containing the cached <code>.pfx</code> file (*)
<code>{4}</code> or <code>{CertFriendlyName}</code>	Friendly name of the newly issued certificate
<code>{5}</code> or <code>{CertThumbprint}</code>	Thumbprint of the newly issued certificate
<code>{7}</code> or <code>{RenewalId}</code>	Id of the renewal
<code>{3}</code> or <code>{StorePath}</code>	Path or store name used by the (first) store plugin
<code>{StoreType}</code>	Name of the (first) store plugin (e.g. CentralSsl or PemFiles)
<code>{OldCertCommonName}</code>	Common name (primary domain name) of the previously issued certificate
<code>{OldCertFriendlyName}</code>	Friendly name of the previously issued certificate
<code>{OldCertThumbprint}</code>	Thumbprint of the previously issued certificate
<code>{vault://json/mysecret}</code>	Secret from the <code>vault</code>

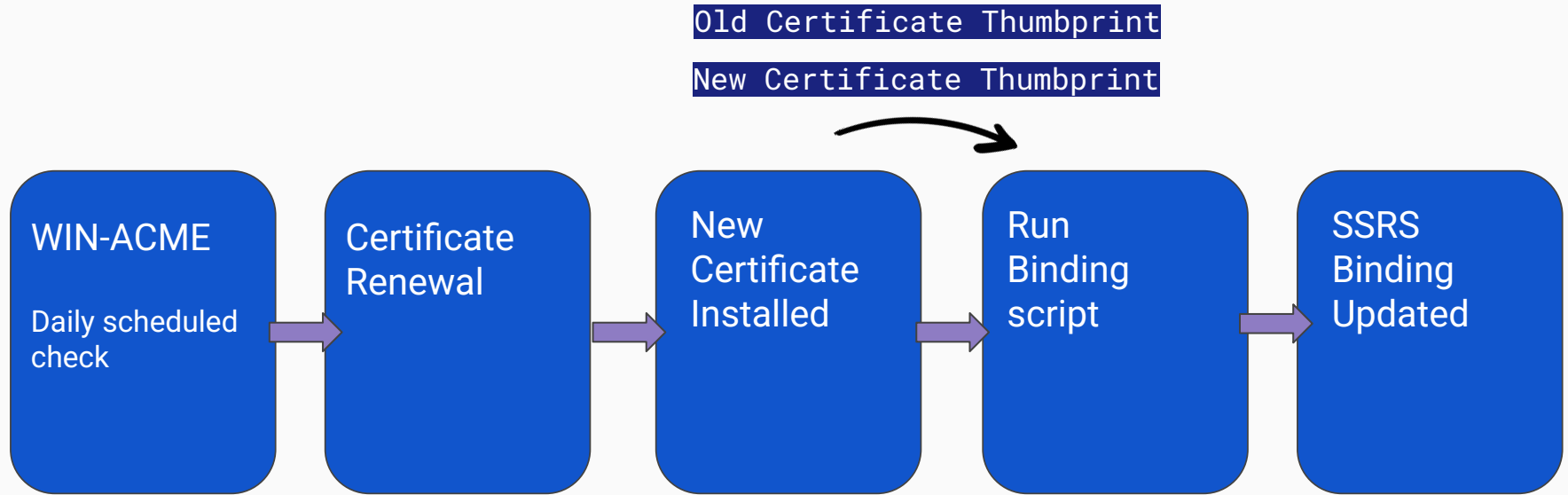
Supports built-in feature to execute external script after certificate renewal

Can pass information about new and old certificates as arguments to the external script

`{CertThumbprint}`

`{OldCertThumbprint}`

WIN-ACME



WIN-ACME External Script Installation

Would you like to store it in another way too?: 5

With the certificate saved to the store(s) of your choice, you may choose one or more steps to update your applications, e.g. to configure the new thumbprint, or to update bindings.

- 1: Create or update bindings in IIS
- 2: Start external script or program
- 3: No (additional) installation steps

Which installation step should run first?: 2

Description: Path to script file to run after retrieving the certificate. This may be any executable file or a Powershell (.ps1) script.

File: C:\Users\anany\Downloads\UpdateSSRSBinding.ps1

[CertCommonName]: Common name (primary domain name)
[CachePassword]: .pfx password
[CacheFile]: .pfx full path
[CertFriendlyName]: Certificate friendly name
[CertThumbprint]: Certificate thumbprint
[StoreType]: Type of store (e.g. CentralSsl, CertificateStore, PemFiles, ...)
[StorePath]: Path to the store
[RenewalId]: Renewal identifier
[OldCertCommonName]: Common name (primary domain name) of the previously issued certificate
[OldCertFriendlyName]: Friendly name of the previously issued certificate
[OldCertThumbprint]: Thumbprint of the previously issued certificate
[vault://json/mysecret]: Secret from the vault

Description: Parameters for the script to run after retrieving the certificate. Refer to <https://win-acme.com/reference/plugins/installation/script> for further instructions.

Parameters: -NewThumbprint {CertThumbprint}, -OldThumbprint {OldCertThumbprint}

Configure External Script

Path to external script

Input parameters
passed to script

WIN-ACME

PROS

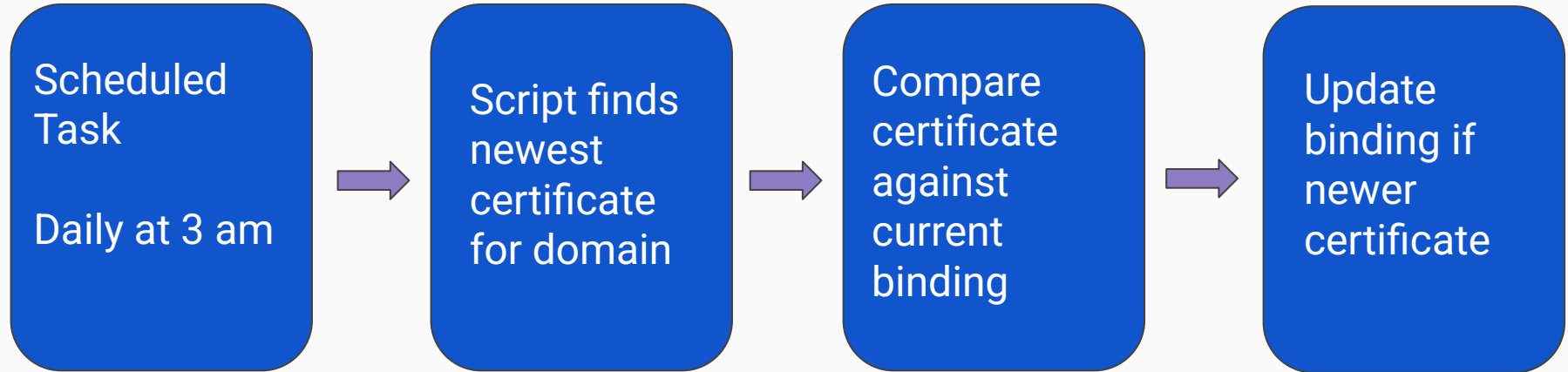
- Fully automated end-to-end
- Triggered exactly when renewal occurs (no delay between renewal and binding)
- Passes certificates' thumbprints directly, no need to search certificate store
- Renewal and binding are coupled as a single unit

CONS

- Requires win-acme to be used for certificate renewals
- Approach couldn't be fully tested for this investigation as it requires a public domain
- If external script fails silently, no built-in retry. Must be monitored and rectified by logging

Alternate Approach: Windows Scheduled Task

Runs the PowerShell script independently, on a fixed timer (e.g daily)



Windows Scheduled Task

PROS

- No dependency on WIN-ACME
- Simple set up with built-in Windows tools (no extra software)
- Easy to inspect and modify (through Task Scheduler GUI)

CONS

- Not event-driven, could create gap between certificate renewal and binding
- Redundant runs if certificate hasn't changed
- No direct link between renewal and binding
- Requires separate maintenance of task schedules

Recommendation



	WIN-ACME (Post Renewal Hook)	Windows Scheduled Task
Trigger	When renewal occurs	Fixed timer
Coupling to Renewal	Tightly coupled (same event unit)	Decoupled (independent of renewals)
Certificate Identification	Thumbprint passed directly	Must search for certificate in store
Setup Complexity	Configured within WIN-ACME's existing menu	Separate task registration
Redundant Runs	None	Highly possible
Dependency	Requires WIN-ACME as renewal tool	Renewal tool independent
Risk of Time Gap	None	Window between renewal and next run

References

1. <https://learn.microsoft.com/en-us/sql/reporting-services/security/configure-ssl-connections-on-a-native-mode-report-server>
2. <https://www.win-acme.com/reference/plugins/installation/script>
3. [https://learn.microsoft.com/en-us/sql/reporting-services/wmi-provider-library-reference/msreportserver-configuration-setting-methods?view=sql-server-ver17\](https://learn.microsoft.com/en-us/sql/reporting-services/wmi-provider-library-reference/msreportserver-configuration-setting-methods?view=sql-server-ver17)
4. <https://ruiromanoblog.wordpress.com/2010/05/08/configure-reporting-services-ssl-binding-with-wmi-powershell/>
5. <https://learn.microsoft.com/en-us/windows/win32/taskschd/task-scheduler-start-page>
6. <https://netwrix.com/en/resources/blog/how-to-automate-powershell-scripts-with-task-scheduler/>